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Criticality-driven enhancer-promoter dynamics in *Drosophila* chromosomes

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eLife Assessment

This manuscript uses modeling approaches to provide mechanistic insight into the structural and dynamic properties of enhancer-promoter interactions in Drosophila. Given the interest in this field, this is a timely approach, and the results give **useful** insights by providing predictions about the processivity of cohesin loop extrusion in Drosophila and concluding that the compartmental interaction strength is poised near criticality in the coil-globule phase space. The evidence provided to support some of the conclusions is, however, **incomplete** and would be strengthened by better considering some of the caveats in the data used to constrain the models, such as the use of "homie" genetic elements in the dynamic data. There is insufficient evidence provided for the dynamics being criticality-driven, and in addition, consideration of alternative models would further strengthen the conclusions of the manuscript.

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Abstract

Recent live imaging in Drosophila embryonic nuclei revealed frequent enhancer–promoter (E–P) contacts across megabase-scale distances, challenging classical polymer models. To identify the physical mechanisms enabling such long-range communication, we performed coarse-grained polymer simulations exploring three chromatin organization modes: ideal polymers, loop extrusion, and compartmental segregation. We found that compartmental segregation, when tuned near the coil–globule phase transition, best captured the experimentally observed structure and dynamics. Adding loop extrusion further improved the agreement with experimental data, suggesting a synergistic interplay. These results indicate that Drosophila chromatin folds near a critical point, enabling dynamic E–P interactions over long distances. Our findings provide a mechanistic framework for chromatin architecture during development and point to criticality as a potentially universal principle of genome organization.

Introduction

The three-dimensional organization and dynamics of chromosomes are key regulators of gene expression in eukaryotes. Chromatin is hierarchically structured within the nucleus, from megabase-scale active and inactive (A/B) compartments to sub-megabase topologically associating domains (TADs) [1–6]. These features arise through distinct molecular mechanisms.

A/B compartments are strongly associated with epigenetic states and transcriptional activity, yet their formation appears to involve multiple, possibly redundant, processes that lead to phase separation of the different chromatin types [7, 8]. In both mammals (e.g., *Mus musculus*) and insects (e.g., *Drosophila melanogaster*), polymer models that incorporate preferential interactions between regions with similar epigenetic marks can robustly recapitulate compartmentalization patterns [9, 10].

While compartmentalization mechanisms appear broadly conserved, the formation of TADs differs between species. In *Drosophila*, TADs likely emerge from the same mechanisms that drive compartmentalization [10]. In contrast, mammalian TADs are primarily shaped by loop extrusion [11–13], a process in which Structural Maintenance of Chromosomes (SMC) complexes such as cohesin extrude chromatin loops until halted by convergently oriented CTCF binding sites, where loops are stabilized [14–16]. Experimental and computational studies support the coexistence of loop extrusion and compartmental segregation in mammals, with loop extrusion dominating at sub-megabase scales and compartmentalization prevailing at larger genomic distances [17–19].

In contrast, the evidence for loop extrusion as a mechanism of chromatin organization in *Drosophila* remains debated [20, 21]. In flies, insulating regions are often found enriched at TAD borders, but they appear to have locus-specific roles [22] and TAD borders are characterized by transitions between epigenetic states rather than by preferentially-bound CTCF [4, 23, 24]. Thus, an alternative mechanistic hypothesis proposes that the formation of TADs in *Drosophila* is likely driven by specific chromatin interactions based on epigenetic marks [21].

The three-dimensional (3D) organization of the genome plays a critical role in cell-type-specific gene regulation during development [25–27]. TADs often encompass genes and their regulatory elements, such as enhancers, facilitating their spatial proximity to target promoters while insulating them from non-target interactions [28–30]. This view of TADs as regulatory domains has helped explain how boundary disruptions can lead to ectopic gene expression [31–33], though exceptions have been described [34, 35]. To better understand the regulatory role of TADs, single-cell and time-resolved approaches have been instrumental. These studies suggest that TADs and compartments represent statistical rather than fixed structural units of genome organization [36], and that the internal dynamics of gene domains are key to modulating promoter–enhancer communication [37].

A recent study by Brückner and Chen *et al.* used live imaging in *Drosophila* embryos to simultaneously track enhancer–promoter (E–P) dynamics and transcriptional activity across genomic distances ranging from s = 58 kb to 3.3 Mb (**Fig. 1A**) [38]. These data enabled inference of chromatin dynamics, which were then compared to predictions from polymer physics models. As expected, the motion of individual loci followed subdiffusive behavior consistent with the Rouse polymer model, in line with previous observations in other eukaryotes [39–41]. However, the two-locus mean-squared displacement deviated markedly from model predictions. Surprisingly, loci separated by large genomic distances exhibited contact frequencies significantly higher than expected from standard polymer models [38]. These findings suggest that long-range enhancer–promoter interactions may be more prevalent than previously thought, but the underlying mechanisms remain unclear. In this study, we aim to address this gap by exploring potential physical and biological drivers of these unexpected long-range contacts.

We modeled and simulated the dynamics of a large chromosomal region *in silico*, incorporating two well-established genome folding mechanisms – compartmental segregation and loop extrusion. We compared our simulation results to available experimental results including: enhancer–promoter (E-P) dynamics and physical distance, as well as Hi-C contact maps. We demonstrate that a simple heteropolymer model can reproduce all experimental data with a good agreement. Combining this model with loop extrusion further refines this agreement, suggesting the possible co-existence of these two mechanisms in *Drosophila*. Notably, the best agreement between simulations and experiments occurs near the critical coil–globule phase transition, where the chromatin fluctuates between extended and compact configurations [42, 43]. These findings suggest that compartmental segregation operating near this transition provides a mechanistic basis for the observed structure and dynamics of *Drosophila* chromosomes.

Results

Polymer simulations are widely used alongside experiments to investigate the structure and dynamics of eukaryotic chromosomes [44–47]. Here, we simulated a large region of chromosome 2R in *Drosophila* (**Fig. 1A**) to reproduce the experimental observations reported by Brückner *et al*. Specifically, we implemented three models: an ideal chain, a block copolymer, and a loop-extruded polymer.

We compared simulation outputs to experimental data using three observables: (i) the Hi-C contact frequency map from the same developmental stage (nuclear cycle 14) [48] (**Fig. 1B**, right), (ii) the mean enhancer–promoter (E–P) physical distance $R(s)$ as a function of genomic separation s (**Fig. 1C**), and (iii) the two-locus mean-squared displacement $\mathrm{MSD}_2(t)$ for varying s (**Fig. 1C**) [38] (see Methods). The first two observables capture structural properties, while the third reflects chromatin dynamics.

To evaluate model performance, we quantified deviations between simulations and experiments using three metrics: $\sigma_{\text{Hi-C}}$ for the Hi-C map, σ_{str} for $R(s)$, and σ_{dyn} for $\mathrm{MSD}_2(t)$ (see Methods). For each model, we identified the parameter set minimizing these deviations and visually assessed the fits to determine consistency with experimental data.

Ideal chain model

We first applied our approach to the simulation of an ideal chain, a basic polymer model with well-established theoretical predictions. A 4.2 Mb segment of chromosome 2R was modeled as a bead-spring polymer without excluded volume (**Fig. 1D**) and simulated as described in Methods. The observed scaling exponents for $R(s)$ and $MSD_2(t)$ were both close to 0.5 (**Fig. 1F,G**), consistent with ideal chain behavior [49].

We then scaled both time and spatial units to match experimental data for the three observables and computed the deviations σ between simulation and experiment. The simulated Hi-C map, using a contact threshold of k = 5 bead diameters (simulation spatial units, s.u.), showed a decay in contact frequency with increasing genomic distance s, but lacked the structured patterns observed in the experimental map ($\sigma_{\text{Hi-C}}$ = 0.982; **Fig. 1E**). The mean spatial distance $R(s)$ deviated from the experimental trend (σ_{str} = 0.796; **Fig. 1F**), and while the fit for $MSD_2(t)$ was reasonable for genomic separations up to 190 kb, it diverged at larger distances (σ_{dyn} = 3.204; **Fig. 1G**), highlighting the limitations of the ideal chain in capturing long-range chromatin behavior.

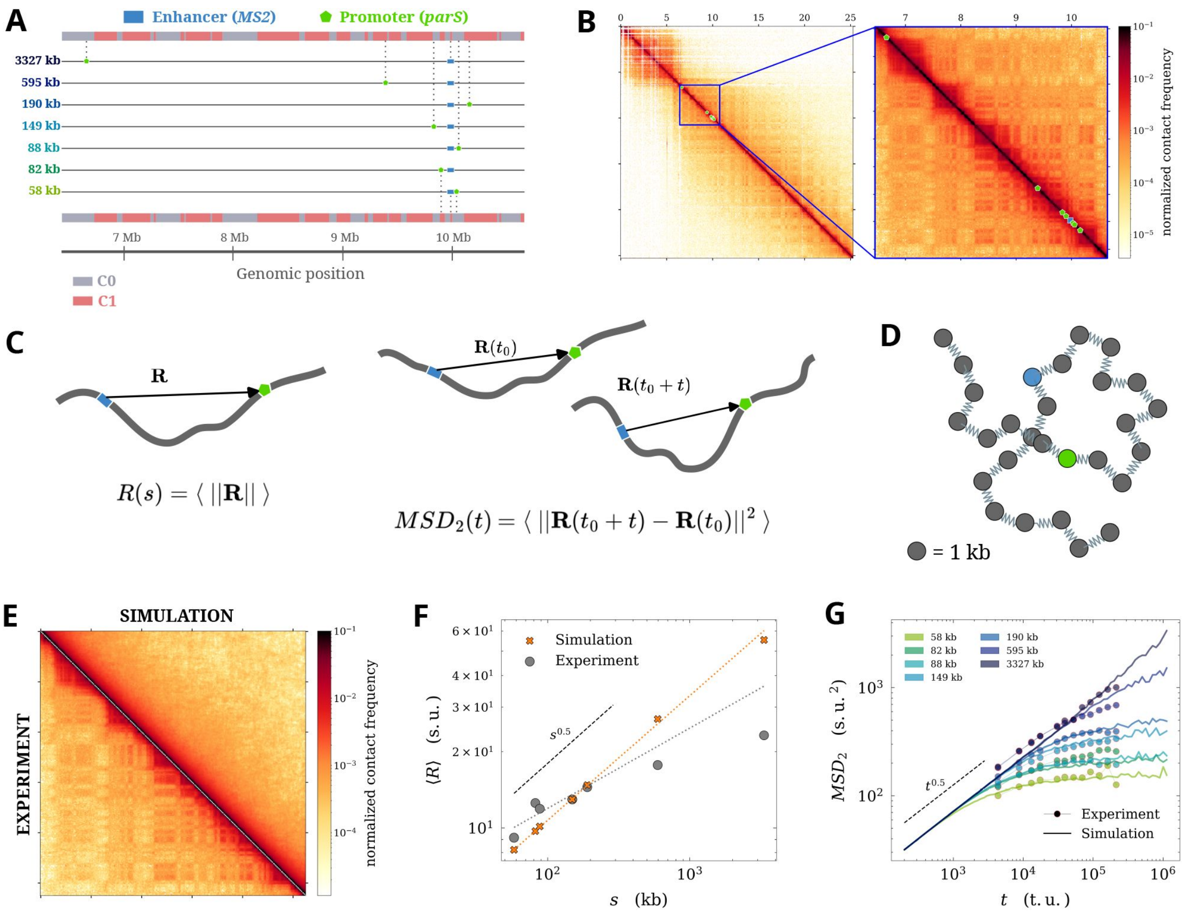

Figure 1

A – Chromosomal region modeled in this study showing insertion sites of the *MS2* (enhancer, blue rectangle) and the *parS* (promoter, green pentagon) from the seven fly lines of [38]. The compartments C0/C1 into which they fall were computed from the contact map in B using eigenvector analysis (Methods). **B** – Hi-C contact map of chromosome 2R (left, from [48]) with the region modeled in the present study enlarged (right). The enhancer and promoter sites use the same markers as in A. **C** – Schematic of the computation of the two observables from [38] used to assess consistency with experimental results. On the left, the mean spatial distance between loci $R(s)$ and on the right, the two-locus mean squared displacement $MSD_2(t)$. **D** Schematic of an ideal chain, i.e., a bead-spring polymer chain without excluded volume. For illustrative purposes, the beads corresponding to the enhancer and promoter site are marked in blue and green, respectively. **E** – Comparison between the Hi-C maps from the experiment and an ideal chain simulation. **F** – Fit for $R(s)$ between the ideal chain simulation and the experiment. Theoretical behavior for an ideal chain indicated by the black dashed line. Gray dotted line shows the linear fit to experimental data (slope $\beta = 0.31$) and the orange dotted line to that of the simulation ($\beta = 0.49$) **G** – Fit for $MSD_2(t)$ for the seven values of genomic distance s between the simulation and the experiment. The black dashed line shows theoretical power law behavior

To test whether excluded volume effects could improve the model, we added steric interactions to the ideal chain. However, this modification further increased the discrepancies with experimental data ($\sigma_{\text{Hi-C}}$ = 0.989, σ_{str} = 1.029, σ_{dyn} = 4.863), as confirmed by visual inspection of the fits (**Fig. S1A–C**). These results indicate that the ideal chain, even with excluded volume, fails to capture the structural and dynamic features observed in *Drosophila* embryos, motivating the exploration of more complex models.

Loop extrusion model

Extensive simulation-based and experimental studies have shown that loop extrusion by chromatin-associated factors—such as cohesins—affects both chromatin compaction and its diffusive scaling behavior [50–52]. We hypothesized that loop extrusion could promote globular chromatin organization while preserving near-ideal chain dynamics. To test this, we implemented loop extrusion using a well-established two-step framework [13, 53]: first, the positions of loop extruding factors (LEFs) were determined on a one-dimensional (1D) lattice, and then these loop constraints were applied to a three-dimensional (3D) bead-spring polymer simulation (**Fig. 2A**; see Methods).

We explored a range of LEF occupancies ⟨Ω⟩ from 0.19 to 1.9 LEFs per 100 kb, consistent with previous estimates [13, 54]. LEF dynamics were governed by four parameters: the loading probability p_L (varied to control ⟨Ω⟩), unloading probability p_U = 0.01, sliding probability p_S = 0.5, and crossover probability p_C = 0.5, which allows LEFs to bypass one another along the chromatin fiber (see Methods for details). By fixing p_U and p_S, we maintained a mean loop size of approximately 100 kb, consistent with in vivo estimates in other organisms [41].

Although TAD boundaries in *Drosophila* are often associated with insulator proteins [20], there is no direct evidence that these elements block LEFs in vivo. Therefore, we did not impose boundary constraints in our simulations; LEFs were allowed to move freely unless stalled by collisions with other LEFs, with the possibility of crossover. To monitor polymer compaction across simulations with varying LEF occupancy, we measured the mean-squared radius of gyration R_g^2 for all enhancer–promoter segments (**Fig. 2B**). As expected, R_g^2 decreased with increasing mean LEF occupancy, indicating progressive chromatin compaction.

We then evaluated how each model's agreement with experimental data changed with LEF occupancy by tracking the three deviation metrics (**Fig. 2C**). The structural deviation σ_{str} decreased with increasing ⟨Ω⟩ and reached a minimum at 1.9 LEFs per 100 kb. In contrast, the dynamic deviation σ_{dyn} was minimized at an intermediate occupancy of ⟨Ω⟩ = 1.14 LEFs per 100 kb. The Hi-C deviation $\sigma_{\text{Hi-C}}$ remained relatively constant across the tested range. These results suggest that loop extrusion contributes to chromatin compaction and can partially reconcile structural and dynamic features observed in vivo, though no single occupancy value optimally fits all observables.

The Hi-C map obtained from the simulation with the lowest deviation revealed a lack of long-range contacts, in contrast to the experimental map (**Fig. 2D**). The fit to $R(s)$ was improved relative to the ideal chain, and the experimental scaling exponent was reproduced up to s = 190 kb. However, $R(s)$ increased markedly at larger genomic distances, reflecting the limited influence of loop extrusion at these scales in our simulations (**Fig. 2E**). Similarly, the model accurately captured the $MSD_2(t)$ curves for s = 58 to 190 kb, but failed to reproduce the rapid plateauing observed experimentally for s = 595 and s = 3327 kb (**Fig. 2F**), indicating incomplete agreement with the dynamic data.

By fitting the scaling factors to convert simulation units to experimental spatial and temporal units (s.u. and t.u., respectively), we estimated a mean LEF sliding speed of approximately 0.8 kb/s, consistent with in vitro measurements of cohesin velocity [55]. The LEF unloading probability

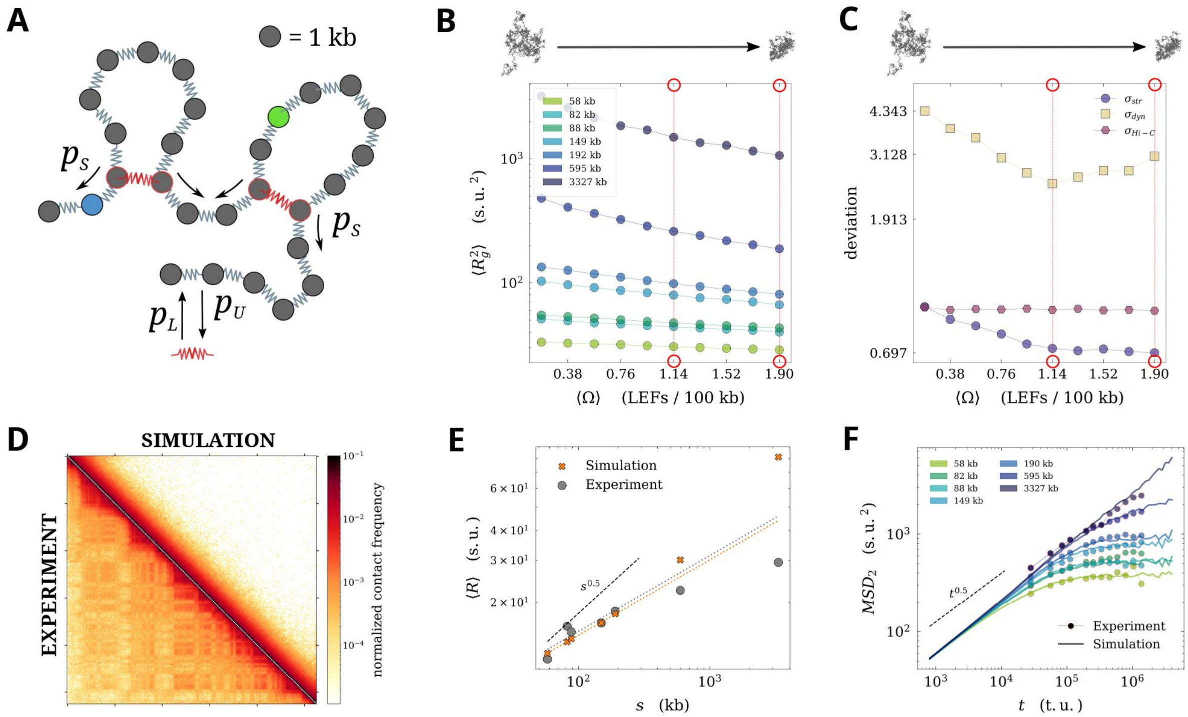

Figure 2

A – Positions of LEFs from the lattice-based stochastic simulation (springs marked in red) are integrated into the bead-spring polymer chain with excluded volume (Methods). LEF activity is controlled by the parameters p_U, p_L, and p_S, which correspond to LEF unloading, loading, and stepping probabilities, respectively. **B** – Mean-squared radius of gyration $(\langle R_g^2 \rangle)$ for each enhancer-promoter segment under increasing mean LEF occupancy ⟨Ω⟩ (in numbers per 100 kb). Red circles and vertical dotted lines indicate the ⟨Ω⟩ at which the best fits to experimental data occur (see inset C). **C** – Deviations between experiment and simulation for *R*(*s*), $MSD_2(t)$ and Hi-C as a function of increasing ⟨Ω⟩. Minima marked by red circles and dotted lines. **D** – Comparison of Hi-C maps between experiment and best fitting simulation for Hi-C. **E** – Best fit of *R*(*s*) between simulation and experiment. The black dashed line indicates ideal chain behavior. Orange and gray dotted lines show linear fits to simulated and experimental data, respectively ($\beta = 0.32$ for the simulation). **F** – Best fit of $MSD_2(t)$ between simulation and experiment. The black dashed line shows the ideal chain exponent.

p_U corresponded to a mean residence time of ~ 63 s, which is substantially shorter than experimental estimates of ~ 360 s obtained under CTCF-depleted conditions in mouse embryonic stem cells [41, 56].

Taken together, these results suggest that while loop extrusion contributes to chromatin organization, it is insufficient on its own to account for all structural and dynamic features observed in the experimental data.

Chromosome structure and dynamics driven by compartments

Next, we investigated the effect of compartmental segregation on chromatin structure and dynamics. To identify chromatin compartments, we computed the first principal component (PC1) of the experimental Hi-C map at 16 kb resolution (**Fig. 1B**, left), assigning each genomic bin to one of two compartments, C0 or C1 (see Methods). This analysis revealed alternating compartment blocks of varying genomic lengths within our region of interest (**Fig. 1A**).

We then assigned each bead in our polymer model to the compartment corresponding to its genomic position (**Fig. 3A**, left). To model compartmental interactions, we introduced a distance-dependent attractive potential between beads belonging to the same compartment, following established block copolymer approaches [10, 45] (**Fig. 3A**, right). For simplicity, we assumed a uniform interaction energy ε for both C0–C0 and C1–C1 bead pairs.

While systematically varying the strength of the attractive potential ε in our simulations, we observed that each enhancer-promoter segment underwent a characteristic coil to globule transition (i.e., larger to smaller mean R_g^2, **Fig. 3B**) [42], consistent with previous studies of self attracting polymers [43, 57]. To identify the value(s) of ε that best matched the experimental data, we computed the deviations σ_{str}, σ_{dyn}, and $\sigma_{\text{Hi-C}}$ across a range of ε values. Both σ_{str} and σ_{dyn} reached minima at $\varepsilon = 0.29$, while $\sigma_{\text{Hi-C}}$ was minimized at a slightly lower value of $\varepsilon = 0.275$ (**Fig. 3C**). Since the contact threshold used to compute Hi-C maps from simulations is not precisely known and depends on experimental crosslinking conditions, we tested a reduced threshold of $k = 4$ s.u. (from $k = 5$ s.u.). This adjustment shifted the minimum of $\sigma_{\text{Hi-C}}$ to $\varepsilon = 0.285$, aligning more closely with the minima of the other two metrics. Visual inspection of the contact maps revealed a strong compartmentalization at both values ε (**Fig. S2A,B**). Overall, this model yielded significantly better agreement with experimental data than previous models ($\sigma_{\text{Hi-C}} = 0.552$, $\sigma_{\text{str}} = 0.314$, $\sigma_{\text{dyn}} = 2.544$).

To further validate this improvement, we visualized the fits to the three observables. Since compartment annotations were derived from experimental data, the simulated Hi-C map closely resembled the experimental one, though with generally lower inter-compartment contact frequencies (**Fig. 3D**). The $R(s)$ curve from simulations followed the experimental trend across all scales (**Fig. 3E**), although the scaling exponent for $s = 58$ to 190 kb was approximately 0.5, differing from the experimental value. The fit to $MSD_2(t)$ was notably improved, with the curves for $s = 595$ and $s = 3327$ kb correctly transitioning into the plateau, although discrepancies remained at shorter time scales for smaller s values (**Fig. 3F**).

Given the overall improvement of the compartmental segregation model over previously tested models, we extended our analysis to a fourth observable from Brückner and Chen *et al.*: the scaling of the *relaxation time* $\tau(s)$ of chromosomal segments. The relaxation time $\tau(s)$ characterizes the timescale at which the two-locus mean-squared displacement $MSD_2(t)$ reaches its plateau, and was computed as described in Methods. For an ideal chain, theory predicts $\tau(s) \sim s^2$, whereas experimental data yielded a scaling exponent of 0.7 ± 0.2 [38]. From our best-fit simulation to dynamics (i.e., the one minimizing σ_{dyn}), we measured a scaling exponent of $\tau(s) \sim s^{0.88}$, which falls within the experimental uncertainty (**Fig. 3G**).

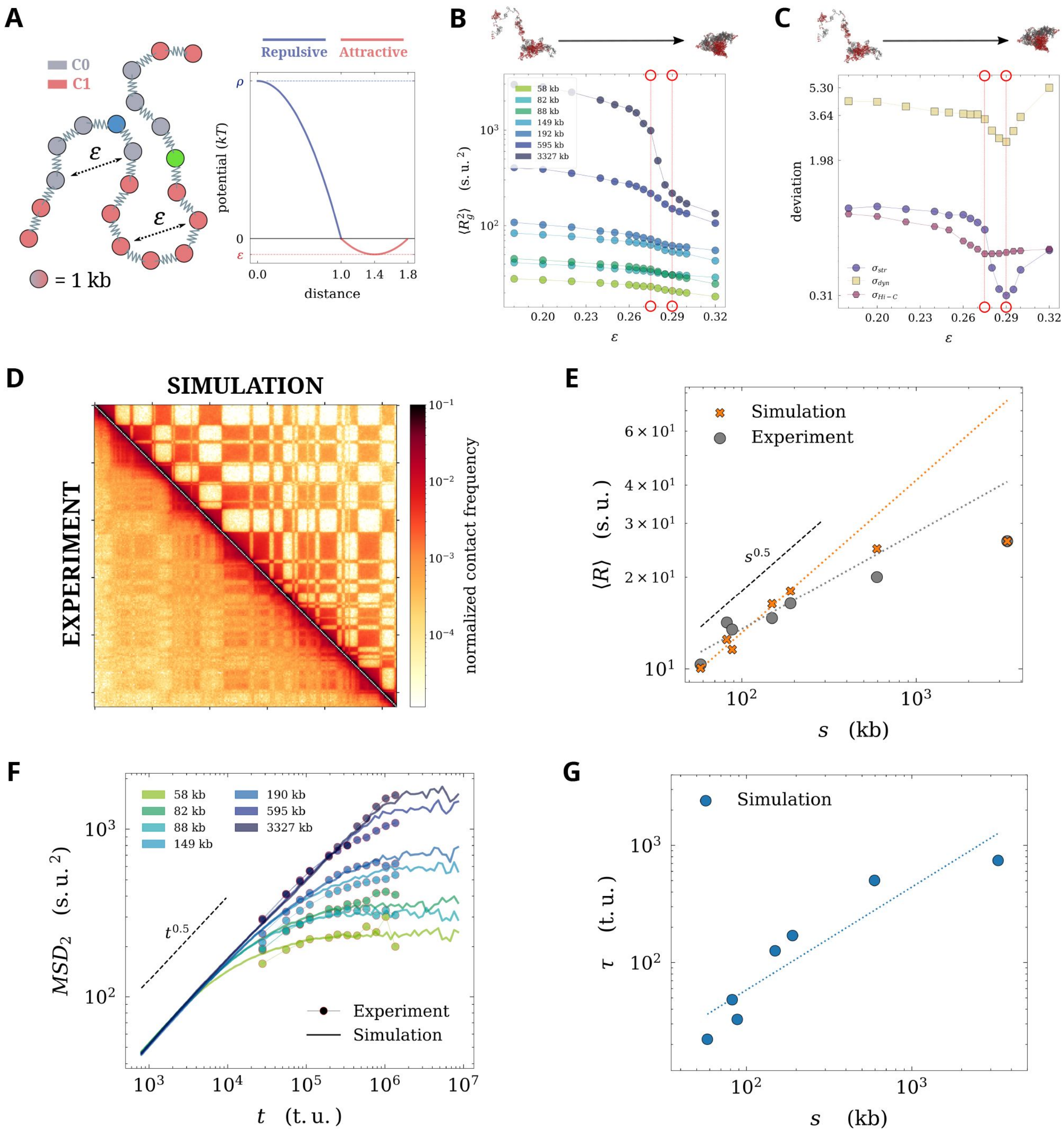

Figure 3

A – (Left) Bead-spring block copolymer chain with excluded volume and two bead types colored based on the compartment in which they belong. The beads corresponding to the enhancer and promoter site are marked as before. (Right) The short-range attractive potential between beads of the same type is given by the parabolic well of depth ϵ (in red), while the repulsive interactions enforce excluded volume (in blue) for all beads. Interactions between beads of different types are only repulsive. **B** – Radius of gyration of each enhancer-promoter segment as a function of compartmentalization strength ϵ. Values of ϵ where best fits with experimental data occur (from following inset with the experimental data) are indicated like in 2. **C** – $\sigma_{Hi\text{-}C}$, σ_{str}, and σ_{dyn} as a function of ϵ. The position of occurrence of best fits is marked in red. **D** – comparison of Hi-C contact maps between simulation and experiment for the best fit to Hi-C. **E** – best fit with experimental $R(s)$ with dotted lines indicating linear fits ($\beta = 0.52$ for the simulated data). The black dashed line shows ideal chain slope. **F** – best fit with experimental $MSD_2(t)$. Black dashed line indicated same as before. **G** – relaxation time computed from the best-fit simulation to $R(s)$ and $MSD_2(t)$. Dashed line indicates scaling exponent $\tau \sim 0.88$

Taken together, our results show that the compartmental segregation model captures multiple experimental observables—including structure, dynamics, and relaxation behavior—more accurately than the ideal chain or loop extrusion models. We therefore conclude that compartmental segregation is a major contributor to the structural and dynamic organization of chromatin in the *Drosophila* embryo.

Combined compartmental segregation and loop extrusion

Genome compartments emerged as the principal driver of chromatin structure and dynamics in *Drosophila*. However, discrepancies remained, particularly at shorter genomic distances and time scales. We hypothesized that additional mechanisms could influence chromatin behavior at these scales. Given its dynamic nature and known interplay with compartmental segregation in mammals—both synergistic and competitive [18, 19, 58]—we considered loop extrusion as a complementary mechanism. Notably, our earlier loop extrusion simulations showed good agreement with $R(s)$ and $MSD_2(t)$ at ~100 kb scales, with deviations primarily at larger distances.

We therefore combined compartmental segregation and loop extrusion in a unified bead-spring polymer model with excluded volume (**Fig. 4A**). To identify optimal parameter combinations, we performed a grid search over a restricted range of values: $0.275 \leq \varepsilon \leq 0.300$ for compartment strength and $0.19 \leq \langle\Omega\rangle \leq 0.95$ LEFs per 100 kb for loop extrusion. For each parameter pair, we computed the deviations from experimental data for all three observables.

The fit to Hi-C favored low LEF occupancy and slightly reduced compartment strength relative to the best-fit compartment-only model, with a minimum at $\varepsilon = 0.280$ and $\langle\Omega\rangle = 0.19$ LEFs/100 kb (**Fig. S3A**). In contrast, the best fit for $R(s)$ occurred at $\varepsilon = 0.300$ and $\langle\Omega\rangle$ between 0.57 and 0.76 LEFs/100 kb (**Fig. S3B**). The dynamic observable $MSD_2(t)$ followed a similar trend, with a minimum at $\varepsilon = 0.290$ and $\langle\Omega\rangle = 0.76$ LEFs/100 kb (**Fig. S3C**). Each deviation metric showed improved minima compared to the compartment-only model.

To identify a global optimum, we normalized the deviations (see Methods) and visualized the combined deviation landscape across the parameter grid. The global minimum occurred at $\varepsilon = 0.290$ and $\langle\Omega\rangle = 0.76$ LEFs/100 kb (**Fig. 4B**), corresponding to the best-fit compartment model supplemented by approximately one freely extruding LEF every 130 kb. We summarized the results in **Fig. S4**, which shows that this combined model provides the best overall agreement with all three experimental observables.

The fit to Hi-C indicated a much higher strength of compartmentalization compared to the experiment in the best-fit simulation case (**Fig. 4C**). $R(s)$ showed overall visual improvement and a significant change to the scaling exponent observed up to $s = 190$ kb ($\beta = 0.41$). This change in the exponent marks a reduction from that of the block copolymer best-fit from the previous section and aligns better with experimental observations (**Fig. 4D**). Lastly, the fit for $MSD_2(t)$ was also in better agreement with experimental data (**Fig. 4E**). The fit at lower s and t showed improvement, the relaxation of the curves for $s = 595$ kb and 3.3 Mb into the plateau was reproduced and we observed better global overlap between the two sets of curves (**Fig. 4E**). Thus, we conclude that the combination of compartmental segregation and loop extrusion displays the best agreement with experimental observations.

Discussion

Here, we leveraged coarse-grained polymer simulations to test three mechanistic models of chromatin organization and reconcile structural and dynamical properties of *Drosophila* chromosomes. First, we confirmed that the ideal chain model, with or without excluded volume, fails to reproduce experimental observations. We then tested a mechanistic model of loop

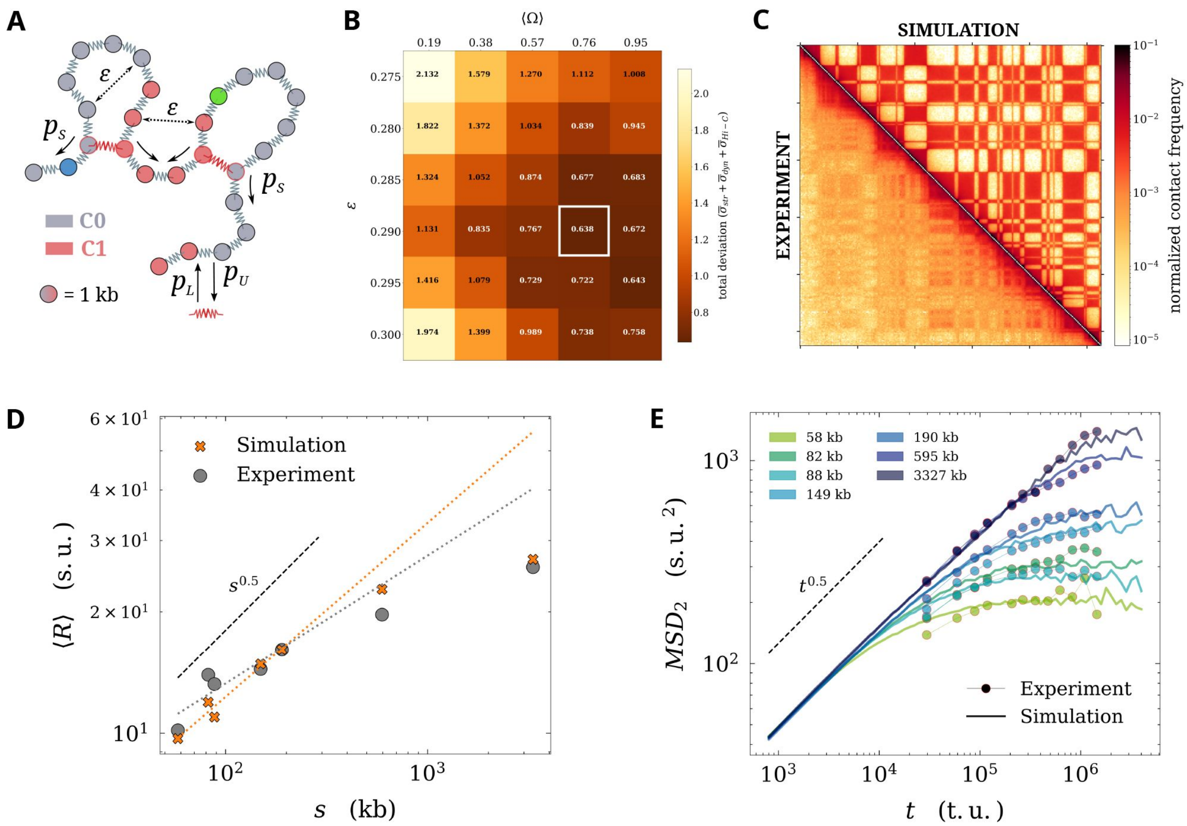

Figure 4

A – Schematic of loop extrusion on a compartment-based block copolymer. ε, p_L, p_U, and p_S are the same as indicated previously. **B** grid with sum of normalized deviations $\overline{\sigma}_{str} + \overline{\sigma}_{dyn} + \overline{\sigma}_{Hi\text{-}C}$ and global minimum highlighted. **C** – Hi-C map comparison between simulation and experiment for global minimum. **D** – fit with experimental $R(s)$ for the same case. Dashed and dotted lines are the same as indicated previously. **E** – best fit with experimental $MSD_2(t)$

extrusion by systematically varying the mean occupancy of loop extruding factors (LEFs) while maintaining their processivity. Although this model captured some features of chromatin dynamics, it was partially inconsistent with structural data. A recent study [52] similarly showed that the two-locus dynamics reported by Brückner and Chen *et al*. could be explained by either fast-sliding LEFs or LEFs with long residence times. However, in the absence of LEF blocking and loop stabilization, loop extrusion alone cannot reproduce the structural features observed in Hi-C maps. Further experimental work is needed to clarify the relationship between cohesins and insulators in *Drosophila*.

We next tested compartmental segregation using a block copolymer model and established its central role in shaping chromatin structure and dynamics. The best agreement with experimental data was achieved near the coil–globule phase transition, a regime known to produce polymers with globular structure and fast dynamics characterized by the Rouse exponent [42, 43]. This near-critical state has also been observed in other eukaryotic systems using imaging-based analyses [42, 43]. We propose that this criticality facilitates enhancer–promoter interactions by reducing their dependence on genomic distance and enabling rapid spatial exploration within the nucleus.

Other recent studies have also sought to reconcile chromatin structure and dynamics using diverse modeling approaches [52, 59, 60]. For example, Kadam *et al*. [52] used simulations of a mouse embryonic stem cell locus to show that loop extrusion could explain both Hi-C and live imaging data. Shi *et al*. [60] used polymer models to generate structural ensembles from contact maps and analyzed them to interpret the results of Brückner and Chen *et al*. Our work complements these efforts by offering a mechanistic interpretation of *Drosophila* chromatin organization based on physical modeling.

Finally, our combined model of compartmental segregation and loop extrusion revealed that compartments dominate genome organization at 100 kb to Mb scales, while loop extrusion may act at shorter length scales. Although the role of loop extrusion in *Drosophila* remains debated [20, 21], our results—together with the conservation of cohesin and CTCF—suggest that chromatin extrusion may indeed occur in flies and contribute to their chromatin architecture.

Conclusion

In this study, we used coarse-grained polymer simulations to systematically evaluate mechanistic models of chromatin folding and reconcile structural and dynamical properties of a *Drosophila* chromosome. Our results showed that neither an ideal chain nor loop extrusion alone could simultaneously reproduce all experimental observables. In contrast, a block copolymer model incorporating experimentally derived genome compartments, particularly near the coil–globule phase transition, significantly improved agreement with experimental data. This agreement was further enhanced by combining compartmental segregation with loop extrusion.

Drawing on recent developments in polymer physics, we propose that *Drosophila* chromosomes fold in a near-critical regime, enabling rapid spatial exploration and frequent long-range enhancer–promoter interactions. This criticality may represent a general principle of genome organization. Future work should investigate whether similar near-critical folding regimes operate in mammalian systems and how additional factors—such as boundary elements, epigenetic modifications, and transcriptional activity—modulate chromatin structure and dynamics across scales.

Materials and Methods

Computing observables from the experiments and the simulations

Two of the four observables, i.e., the end-to-end distance $R(s)$ and the two-locus mean-squared displacement $MSD_2(t)$, were computed from the raw trajectory data of Brückner & Chen *et al.* [61] using the code originally published with the article [62]. From the simulation trajectories comprising the coordinates of nearly 7×10^4 polymer conformations sampled over time, $R(s)$ and $MSD_2(t)$ were calculated for s = 58, 82, 88, 149, 195, 595, 3327 kb using Eq. 1 and Eq. 2 respectively.

$$R(s) = \langle \ ||\mathbf{R}|| \ \rangle \tag{1}$$

$$MSD_2(t) = \langle \ \mathbf{R}(t_0 + \Delta t) - \mathbf{R}(t_0) \ \rangle_{t_0} \tag{2}$$

The effect of the localization error in [38, 63] was tested on $R(s)$ from the simulation trajectories to ensure that this factor could not affect the fits significantly. Indeed, the effects of localization error were negligible (**Fig. S5**).

The relaxation time τ (s) was measured from the simulations by finding the intercept between the initial regime and the plateau of each $MSD_2(t)$ curve; the scaling exponent was then estimated by computing the slope of $\tau(s)$ plotted in the logarithmic scale.

The experimental Hi-C map (16 kb resolution) was obtained from another study [48] performed during the same developmental stage of the *Drosophila* embryo as in Brückner & Chen *et al.* The Hi-C map for each simulation was generated by first coarse-graining polymer conformations from the trajectory to 16 kb resolution, then calculating pairwise distances between beads and determining contacts with a contact threshold of k bead radii. These contact maps were then averaged and normalized using Sequential Component Normalization [64]. The contact threshold was selected to ensure consistently high Spearman's correlation [65] between simulated and experimental Hi-C maps across all simulations (i.e., for all values of ε; **Fig. S2C**). This analysis identified a suitable range of thresholds, $5 \leq k \leq 8$ simulation units (s.u.). We selected $k = 5$ as a representative value, as it provided strong correlation and corresponded to a reasonable physical distance after conversion to nanometers (see next section).

Fitting observables between simulations and experiments

To fit the experimental $R(s)$ to the simulated one, the curve was translated along the y-axis (i.e., the axis corresponding to length) in the logarithmic scale (**Eq. 3**, equivalent to scaling by a numerical factor) until the deviation σ_{str} between the experimental and the simulated curves reached a global minimum. No translation was necessary along the x-axis since the genomic distance s was common to both sets of data. The L-BFGS-B optimization algorithm [66, 67] was used for this fit.

$$\sigma_{str} = \sum_{s=58 \text{ kb}}^{3.3 \text{ Mb}} \left| \ R^{sim}(s) - \left(R^{exp}(s) + \Delta_R\right) \ \right| \tag{3}$$

This fit was then used to convert simulation spatial units s.u. (i.e. bead-bead bond length) into real spatial units (i.e. nm) using Eq. 4).

$$l_{exp} \text{ s.u.} = l_{exp} \text{ nm} \times 10^{-\Delta_R} \text{ s.u. nm}^{-1} \tag{4}$$

This correspondence allowed us to convert the contact threshold used to compute Hi-C maps in real spatial units. Depending on the simulation case, the contact threshold k = 5 varied roughly between 132 and 308 nm based on Δ_R computed from the corresponding $R(s)$ fits. In the case of the best-fit for the block copolymer, $10^{-\Delta_R} = 50.02$, and the contact threshold was estimated to be ≈ 250 nm.

Fitting $MSD_2(t)$ required a more elaborate procedure, as the simulated and experimental datasets contained different numbers of points. Each simulated $MSD_2(t)$ curve was first fit to a fifth-degree polynomial $\mathcal{F}_s$ in logarithmic scale to interpolate the data and enable functional evaluation. Then, using the same optimization algorithm, the experimental data points for each genomic separation s were translated along both space and time axes in log–log space—equivalent to fitting both squared displacement and time simultaneously—until the deviation σ_{dyn} was minimized (Eq. 5).

$$\sigma_{dyn} = \sum_{s=58\text{ kb}}^{3.3\text{ Mb}} \left|\ MSD_2^{exp}(t) + \Delta_{MSD_2} - \mathcal{F}_s(t+\Delta_t)\ \right| \tag{5}$$

As with spatial units, real time units (in seconds, s) can be converted to simulation time units (timestep of the Langevin integrator, t.u.) using the temporal scaling factor Δ_t obtained from the fitting procedure (Eq. 6). For the best-fit case of loop extrusion, we obtained $10^{-\Delta_t} = 1278.71$, implying that 1 s = 1278.71 t.u.

$$t_{exp}\ \text{t.u.} = t_{exp}\ \text{s}\ \times 10^{-\Delta_t}\ \text{t.u. s}^{-1} \tag{6}$$

Lastly, to estimate the deviation $\sigma_{Hi\text{-}C}$ between the simulated and the experimental Hi-C map, we compute one minus the Spearman's correlation (ρ) between the corresponding observed-over-expected maps (see Eq. 7).

$$\sigma_{Hi\text{-}C} = 1 - \left|\ \rho\left[\ \mathsf{OE}(\mathsf{SCN}(M_{sim})),\ \mathsf{OE}(\mathsf{SCN}(M_{exp}))\ \right]\ \right| \tag{7}$$

where M_x denotes a Hi-C map, SCN is the normalization function [64], and OE the observed-over-expected function, consisting in calculating for each value the log ratio with an expected value corresponding to the diagonal on which this value is located.

To compute a global minimum for all observables, the individual deviations were normalized and combined into a single metric. Specifically, the global deviation was defined as $\overline{\sigma} = \overline{\sigma_{\text{Hi-C}}} + \overline{\sigma_{\text{str}}} + \overline{\sigma_{\text{dyn}}}$, where each $\overline{\sigma_x}$ represents the min–max normalized deviation for observable x, computed across all parameter combinations for that simulation case.

Modeling and simulations

The 4.2 Mb stretch of chromosome 2R (chr2R:6,457,453–10,658,620), encompassing all enhancer–promoter segments, was modeled as a fully flexible homopolymer bead-spring chain—i.e., an *ideal chain*—with a total of 4208 beads, each representing 1 kb of genomic distance. For simplicity, the *MS2* and *parS* insertions used in the experiment were modeled as point insertions along the polymer.

Based on data from [38], promoter loci were assigned to beads 3563, 3439, 3593, 3372, 3695, 2926, and 193. The enhancer was positioned at bead 3505 for downstream promoter loci and at bead 3521 for upstream ones, preserving the experimental enhancer–promoter genomic separations of 58, 82, 88, 149, 190, 595, and 3327 kb.

The conformational dynamics of this ideal chain are described by the Rouse model [49, 68]. In all models except the Rouse model, excluded volume interactions were introduced using a parabolic repulsive potential, as defined in Eq. 8.

$$E_{rep} = \rho\left(1 - \left(\frac{r}{r_{rep}}\right)^2\right) \tag{8}$$

where the maximum height of the potential was ρ = 5 kT and its width was r_{rep} = 1

For all models, the thermal fluctuations of the polymer chain over time were simulated under over-damped Langevin dynamics [69] using the OpenMM-based *polychrom* package [70, 71]. The simulations of the Rouse model with and without excluded volume were first equilibrated for 10^7 t.u., then the chain dynamics were recorded for 10^9 t.u., with one conformation being sampled every 10^3 t.u.. For all other simulations, these values were 8×10^6 t.u., at least 5.6×10^8 t.u. and 8×10^2 t.u. respectively. For the simulations of the Rouse model, the only parameters used were the equilibrium bead-bead harmonic bond length (1 s.u.) and its standard deviation (≈ 0.28 s.u.); the parameters for the other models are discussed in the sections below.

Loop extrusion

A common two-step framework was used to simulate loop extrusion on the chromosome [13, 53]. For each case, first, a stochastic simulation of LEF dynamics was performed on a one-dimensional lattice of the same length as the polymer. LEFs could stochastically load onto and unload from the lattice. They were allowed to slide across the lattice, and while doing so, could cross one another or collide and stall. The loading, unloading, sliding, and crossing of LEFs were dictated by a set of loop extrusion parameters (**Table 1**). No insulators or boundary elements were introduced in any of the simulation cases, and therefore, the LEFs moved freely unless they were stalled by another LEF. During the simulation, the positions of the LEFs on the lattice were recorded over time to keep track of the loops formed and dissolved. Next, during the 3D Langevin dynamics simulation of the polymer, beads corresponding to the LEF positions from the lattice-based simulation trajectory were brought together and anchored by harmonic bonds to mimic the bridging action of LEFs, thus forming loops in the chain. By constantly moving the loop anchors to beads that matched the positions on the lattice-based trajectory, dynamic loop extrusion was simulated on the polymer.

The stochastic model was implemented such that the ratio between the probabilities of LEF loading p_L and unloading p_U determined the steady state mean occupancy of LEFs ⟨Ω⟩ on the lattice (Eq. The steady state was dependent on the size of the lattice and the sliding probability of the LEFs p_S, and was achieved for time $t >> 10^4$. Therefore, to vary ⟨Ω*⟩*, p_L was varied while keeping the sliding probability p_S and p_U constant. p_S = 0.5 and p_U = 0.01 were assigned to attain a mean LEF p_U processivity $\lambda = 2 \times \frac{p_S}{p_U} \approx 100\ \mathrm{kb}$

$$\frac{d\Omega}{dt} = p_L - p_U\Omega$$

Steady state achieved when $\frac{d\Omega}{dt} = 0$, therefore

$$\langle\Omega\rangle = \frac{p_L}{p_U} \tag{9}$$

From **Eq. 6** and the average time taken for a LEF to slide 1 kb (800 t.u.,1), LEF parameters were estimated in real units. Depending on the simulation case, $1074.69 \leq 10^{-\Delta_t} \leq 1664.39\ \mathrm{t.u.\ s^{-1}}$ taking the best-fit case, $10^{-\Delta_t} = 1278.71\ \mathrm{t.u.\ s^{-1}}$ yielding a mean LEF speed of

Parameter	Value
Lattice length	4208
LEF unloading probability	0.01
LEF loading probability	0.08 to 0.8, stepping 0.08
LEF sliding probability	0.5
LEF crossing probability	0.5
LEF anchor length	1.25 s.u.
Time units for LEF probabilities	800 t.u.

Table 1.

Parameters of loop extrusion for stochastic simulations on the lattice

$p_S \times \frac{10^{-\Delta_t}\ \mathrm{t.u.\ s^{-1}}}{800\ \mathrm{t.u.}} = 0.5 \times \frac{1278.71\ \mathrm{t.u.\ s^{-1}}}{800\ \mathrm{t.u.}} \approx 0.8\ \mathrm{kb\ s^{-1}}$ Similarly, the mean LEF residence time was $\frac{1}{p_U} \times 800\ \mathrm{t.u.} \times 10^{\Delta_t}\ \mathrm{s\ t.u.^{-1}} = \frac{1}{0.01} \times 800\ \mathrm{t.u.} \times 10^{-3.107}\ \mathrm{s\ t.u.^{-1}} \approx 63\ \mathrm{s}$

Compartmental segregation

The compartments of the chromosomal segment were predicted by computing the first principal component of the genomic distance normalized experimental Hi-C map at 16 kb resolution [48]. A large region of the chromosome harboring the loci of interest but excluding the centromere (chr2R:6,464,000– 25,286,936) was used for the computation. Each 16 kb segment of this stretch was associated with one compartment (names C0 or C1). Each bead in the bead-spring chain with excluded volume was assigned a type based on the compartment predicted, resulting in blocks of alternating C0 and C1 beads. Attractive interactions were introduced only between beads of the same type during Langevin dynamics simulations. The strength of attraction between the beads was controlled by the parameter ε, which corresponds to the depth of the attractive well in a custom parabolic potential (Eq. 10). This potential was devised to induce strong attraction between beads at a short range.

$$E_{att} = \varepsilon\Big(\frac{4}{(r_{rep} - r_{att})^2}\Big(r - \frac{r_{rep} + r_{att}}{2}\Big)^2 - 1\Big) \quad (10)$$

where $0.18 \le \varepsilon \le 0.32$ and the width as measured from $r = 0$ was $r_{att} = 1.8$

Supplementary Figures

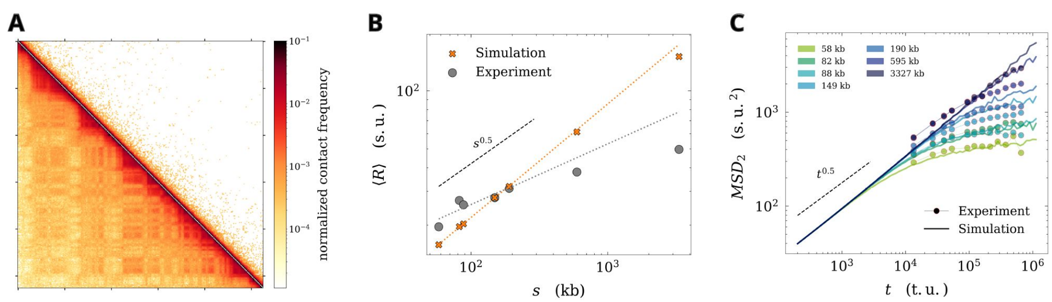

Figure S1

Fits for the Rouse model with excluded volume (i.e., a swollen coil) **A** – Hi-C, **B** – ⟨R⟩, **C** – MSD_2

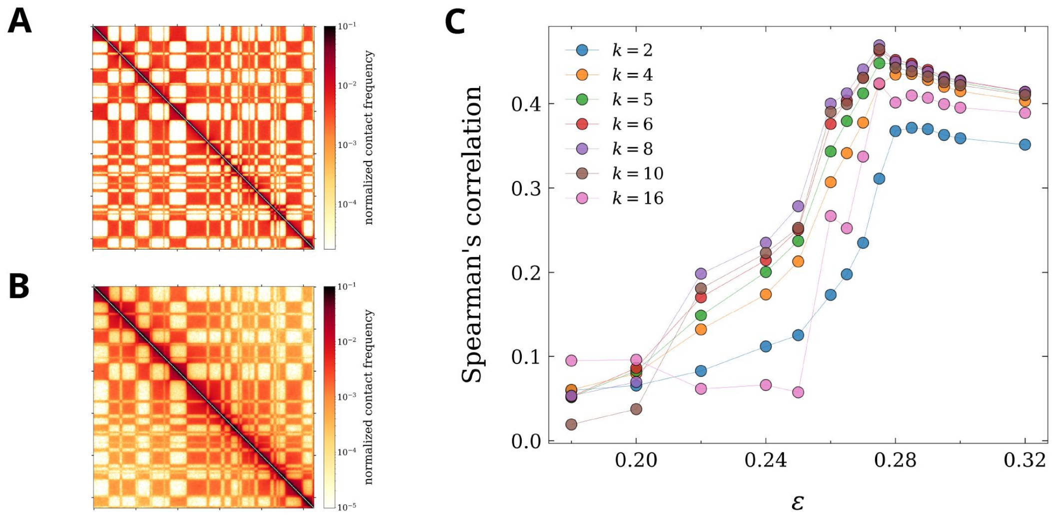

Figure S2

A – Low contact threshold k = 4, best fit at ε = 0.285. **B** – High threshold k = 6, best fit at ε = 0.275. **C** – Choosing a suitable cut-off from comparisons with Hi-C data for the block copolymer simulations.

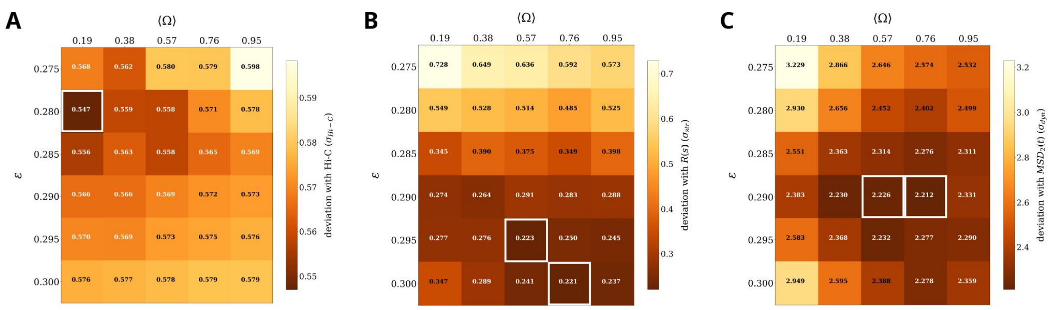

Figure S3

A – Grid search with parameter pairs of ε and ⟨Ω⟩ for the compartment-based block copolymer with loop extrusion showing σ_{str}, two minima highlighted. **B** – grid showing σ_{dyn} for each simulation. Minima highlighted as in inset A. **C** – $\sigma_{Hi\text{-}C}$ as a function of key parameters

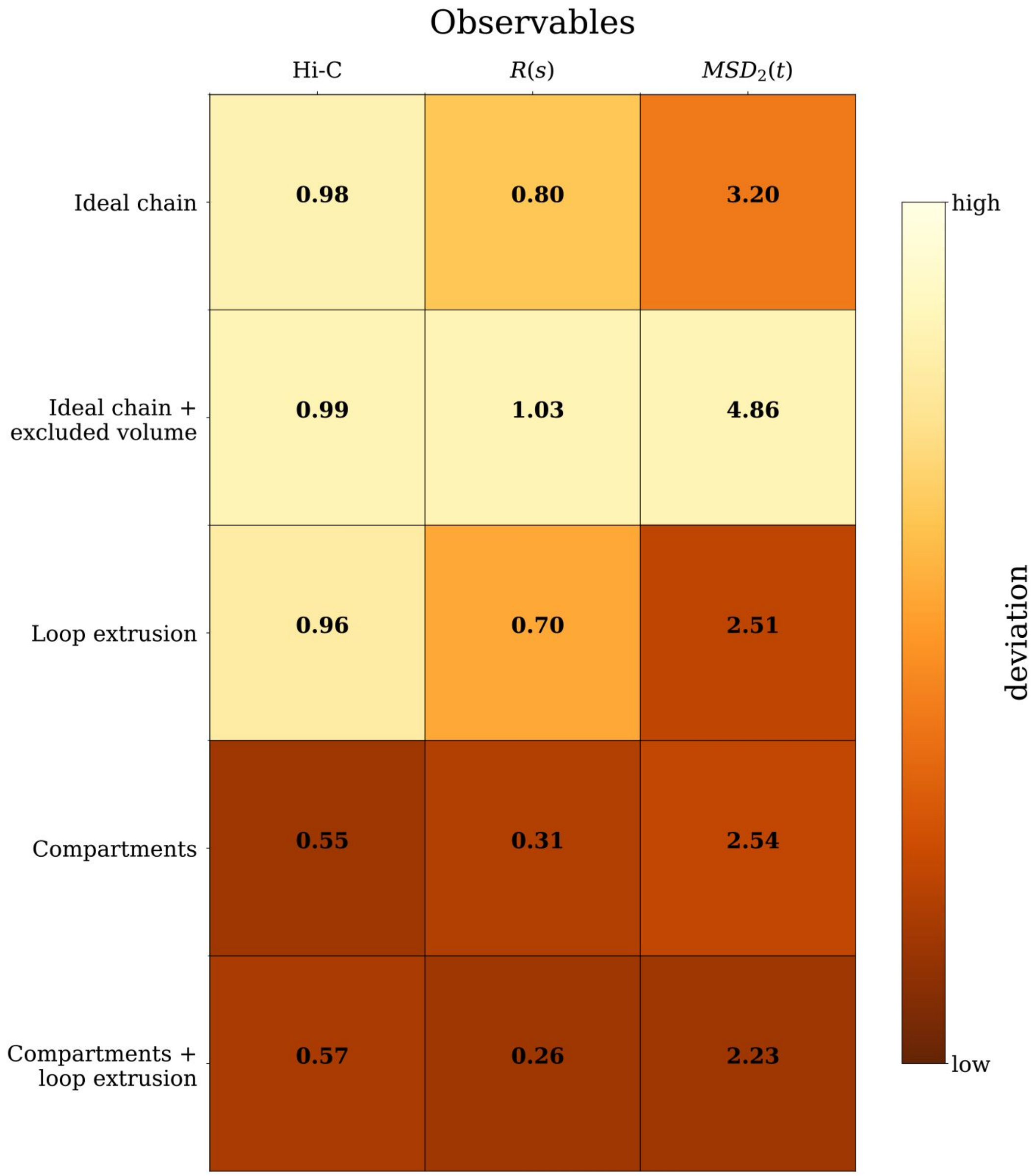

Figure S4

Summary of deviations for all observables: Hi-C (1 - Spearman's correlation), $\langle R \rangle$ (absolute deviation in logarithmic scale), and MSD_2 (same as before) across all explored models

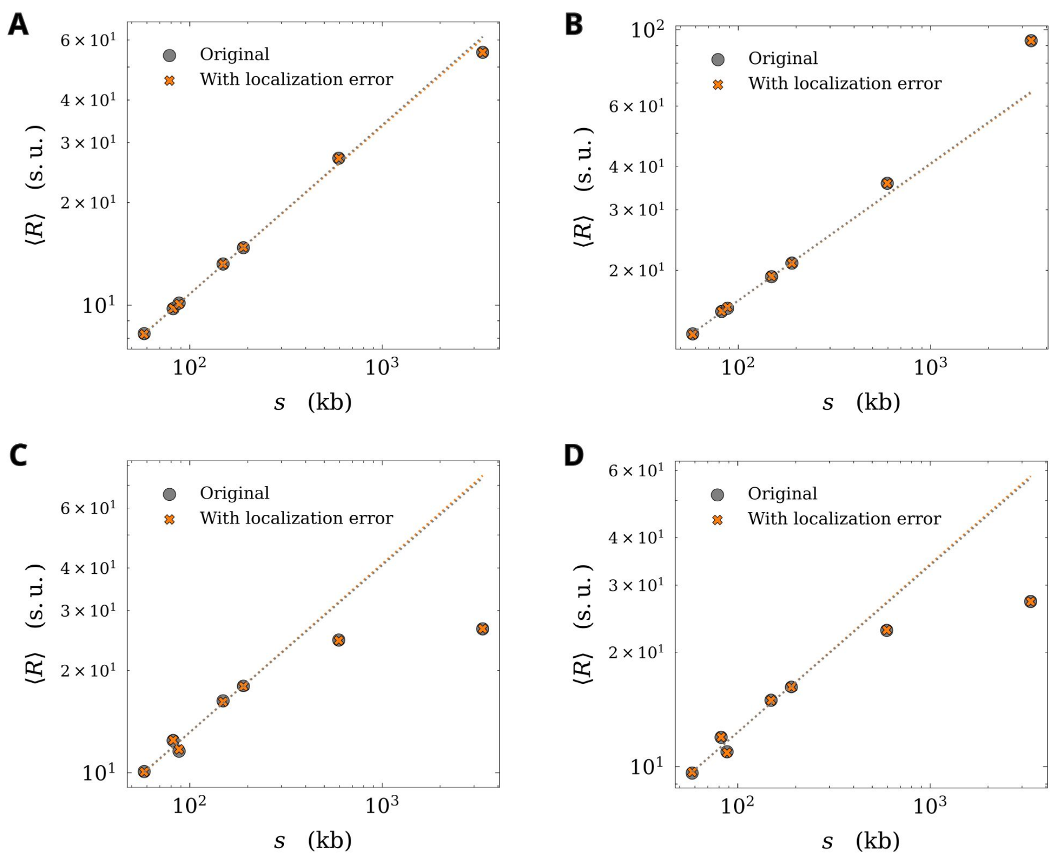

Figure S5

Effect of localization error on end-to-end distance $\langle R \rangle$ is negligible for all simulations, **A** – Rouse model, **B** – Best-fit loop extrusion, **C** – Best-fit compartment-based block copolymer, **D** – Global best-fit for loop extrusion on compartment-based block copolymer

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Editors

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Reviewer #1 (Public review):

Summary:

This computational study investigates the physical mechanisms underlying enhancer-promoter (E-P) interactions across genomic distances in Drosophila chromosomes, motivated by a previously published study that revealed unexpectedly frequent long-range contacts challenging classical polymer models. The authors performed coarse-grained polymer simulations testing three chromatin organization models: ideal polymers, loop extrusion, and compartmental segregation, comparing their predictions to experimental Hi-C contact maps, mean E-P distances, and two-locus mean-squared displacement dynamics. They found that compartmental segregation best captured both the structural and dynamic features observed experimentally, while neither ideal chains nor loop extrusion alone could reproduce all experimental observables. The combination of compartmental segregation with loop extrusion further improved agreement with experimental data, suggesting these mechanisms might be involved in Drosophila chromatin organization.

Strengths:

The paper has two primary strengths:

(1) The simulations are based on biologically interpretable mechanisms (compartmentalization and loop extrusion), which may facilitate making specific experimentally testable predictions.

(2) The work uses a systematic approach to increase model complexity by directly fitting to data, first establishing that simple models fail to capture the data until arriving at a more complex model that does capture the data.

Weaknesses:

I have two major concerns (detailed below) and multiple minor concerns.

Major concerns:

(1) While the upside of the mechanistic simulations is that they are interpretable, the downside is that specific choices for the considered mechanism were made, and conclusions drawn from it are necessarily biased by the initial choices. In this paper, only two mechanisms were considered: loop extrusion and compartmentalization. Yet, it is not clear why these are the most likely underlying mechanisms that might determine the chromosome dynamics. Indeed, previous work (not cited in this paper) showed that Drosophila chromosome structure is not determined by loop extrusion: https://elifesciences.org/articles/94070.

This should be acknowledged, and the main reasons for choosing these particular mechanisms should be laid out. The conclusions of the paper must then necessarily always be seen under the caveat that only these two mechanisms were considered.

(2) Even within the framework of the approach, insufficient evidence is given to support the title of the paper "Criticality-driven enhancer-promoter dynamics in Drosophila chromosomes" for two reasons:

(a) The fact that the best-fit parameters are near a coil-globule transition does not mean that the resulting dynamics are criticality-driven. To claim criticality, one would usually expect much more direct evidence, such as diverging correlation lengths. Furthermore, it would need to be shown that the key features of the dynamics (which should be defined, presumably the static and dynamic exponents) indeed depend on the parameters being at this transition. i.e., when tuning the simulations away from this parameter point, does the behaviour disappear? Only in this case can it be claimed that the behaviour is driven by this phenomenon.

(b) The results section actually contains no mention of the coil-globule transition, and it is not clear in what way the parameters are close to this transition.

Thus, three things are necessary:

(i) How the parameters are close to the transition needs to be explained in detail.

(ii) The divergence of observed dynamics whenever the parameters are tuned away from the transition needs to be demonstrated.

(iii) Even if 1 and 2 are fulfilled, a more careful title should be chosen, such as "Polymer simulations near the coil-globule transition are consistent with enhancer-promoter dynamics in Drosophila chromosomes."

Many of the results in the figures and results section are rather repetitive and could be compressed. The main result of Figure 1 - that the data are not described by an ideal chain - was already fully shown and established in the original paper from which the data are taken. Figure 2 is a negative result with near-identical panels to Figure 3. Figure 4B is hard to interpret.

The paper makes no concrete suggestions for new experiments to test the hypotheses formulated. Since the paper can only claim that the simulations are consistent with the data, it would significantly strengthen the paper if testable predictions could be made.

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Reviewer #2 (Public review):

Summary:

In this work, Ganesh and colleagues use experimental data from Hi-C and from live-cell imaging to evaluate different polymer models of 3D genome organization in Drosophila based on both structural and dynamic properties. The authors consider several leading hypotheses, which are examined sequentially in increasing level of complexity - from the minimal Rouse polymer, to a model combining sequence-specific compartmentalization and loop-extrusion without extrusion blockers. They conclude that the combination of both compartmentalization and loop-extrusion gives the best agreement with the data. Their analysis also leads to concrete predictions about the processivity of cohesin loop extrusion in Drosophila, and a conclusion that the compartmental interaction strength is poised near criticality in the coil-globule phase space.

Strengths:

There is considerable interest in the field in understanding the mechanisms responsible for the 3D spatial organization genome and the dynamic movement of the genome, which has major implications for our understanding of long-range transcriptional regulation and other genome behaviors. The live-cell experimental work on which this study draws highlights the limitations of existing models to explain even the dynamic behaviors observed in the data, further exciting interest in further exploration. Therefore, this paper seeks to address an important gap in the field. The work is written in a well-organized, well-illustrated fashion. The text and figures are nicely integrated, easy to read, and explain challenging concepts with elegance and brevity in a manner that will be accessible to a broad audience.

Weaknesses:

The validity and utility of these conclusions are, in my view, substantially undermined by what appears to be unappreciated peculiarities of the live-cell data set that was used to constrain the model. The live-cell data comes from embryos were edited in a way that intentionally substantively changed both the 3D genome structure and dynamics specifically at the loci which are imaged, a case which is not at all explained by any of the models suggested nor acknowledged in the current work, nor compatible with the Hi-C data that simultaneously used to explain these models. As these ignored synthetic alterations have been previously shown to be determinative of transcriptional activity, the relevance of the author's work to transcriptional control (a prime motivation in the introduction) is unclear.

The agreement in 3D organization, as represented in chromosome-scale contact frequency heatmaps, is substantially less impressive than the agreement seen in prior work with similar models. This discrepancy appears to be due in part to the unappreciated effects of the mentioned in the previous limitation, as well as inappropriate choices in metrics used to evaluate agreement. It is also not particularly surprising that combining more models, with more free parameters, results in an improvement in the quality of fit.

Some major results, including both theoretical works and experimental ones, are ignored, despite their relevance to the stated objective of the work. The current manuscript and analysis could be improved substantially by a consideration of these works.

I describe these issues in more detail below.

Major issues:

(1) The genetic element "homie" is present in a subset of the data: The experimental data used in this analysis come from different fly lines, half of which have been edited explicitly to alter genome structure and consequent transcriptional behavior, yet the authors are trying to fit with a common model - a problem which substantially undermines the utility of the analysis.

Specifically, the authors evaluate the various models/simulations by comparing them to Hi-C from wildtype Drosophila embryos on the chromosome scale and 3D distances and dynamics from live cell imaging in genetically edited embryos, to a series of models in turn. The exercise fatally overlooks a critical fact, (admittedly not easily noticed in the work from Bruckner et al), that the fly embryos used for nearly all their analyses contain not only fluorescent labels, but also contain two copies of a powerful genetic sequence, "homie", known for its ability to dramatically change the 3D organization and dynamics of the genome. Whether or not the fluorescent labels themselves used in the study further alter structure and dynamics is not entirely clear (and will require further work beyond the scope of either study), but at least these fluorescent labels aren't known to dramatically affect 3D structure and dynamics the way homie is. The critical problem is that adding or removing the "homie", as shown in a collection of prior works I describe below in more detail, dramatically affects structure, dynamics, and gene expression. Whether or not the genome contains two distal cis-linked copies of homie fundamentally changes genome structure and dynamics, so to use one dataset which has this edit (the live-cell data) and one dataset which lacks it (the Hi-C data) is, in some sense, to guarantee failure of any model to match all the data.

If the authors had chosen instead to focus exclusively on the 'no homie' genetic lines in the Brukner data, they would have a much smaller dataset (just 2 distances), which would not cover all the length scales of interest, but it would at least be a dataset not known to be contradictory to the Hi-C. The two 'no homie' lines make much more plausible candidates for the sort of generalizable polymer dynamics these authors seek to explain, as will hopefully be made more clear by a brief review of what is known about homie. I next describe the published data that support these conclusions about how homie affects 3D genome spatial organization and dynamics:

What is "homie" and how does it affect 3D genome distances, dynamics, and gene expression?

The genetic element "homie" was named by James Jaynes' lab (Fujioka...Jaynes 2009) in reference to its remarkable "homing" ability - a fascinating and still poorly understood biological observation that some genetic sequences from Drosophila, when cloned on plasmids and reintegrated into the genome with p-elements, had a remarkable propensity to re-integrate near their endogenous sequence, (Hama et al., 1990; Kassis, 2002; Taillebourg and Dura, 1999; Bender and Hudson, 2000; Fujioka...Jaynes 2009). By contrast, most genetic elements tend to incorporate at random across the genome in such assays (with some bias for active chromatin).

The Jaynes lab subsequently showed that flies carrying two copies of homie, one integrated in cis, ~140 kb distal from the endogenous element, formed preferential cis contacts with one another. Indeed, if a promoter and reporter gene were included at this distal integration site, the reporter gene would activate gene expression in the pattern normally seen by the gene, even-skipped. The endogenous copy of homie marks one border of ~16 kb mini-TAD which contains the even-skipped gene, (eve), and its developmental enhancers, so this functional interaction provides further evidence of physical proximity (as was also shown by 3C by Jaynes (Fujioka..., Schedl, Jaynes 2016), and later with elegant live imaging, by Jaynes and Gregor (Chen 2018)).

Critically, if either copy of homie is deleted or substantially mutated, the 3D proximity is lost (Fujioka 2016, Chen 2018, Bruckner 2023), and the expression of the transgene is dramatically reduced (at 58 kb) or lost. Given the author's motivation of understanding "E-P" interactions, the fact that the increased 3D proximity provided by homie is as essential for transcription as the promoter itself at the ~150 kb distance, underscores that these are not negligible changes.

These effects can be seen by plotting the data from Bruckner 2023, which includes data from labels with separations of 58 kb and ~150 kb "no homie" as well as homie. Unfortunately, the authors don't plot this data in the manuscript in the comparison of 3D distances, though the two-point MSD can be seen in Figure S13C, and laudably, the data is made public in a well-annotated repository on Zenodo, noted in the study. Note that the distance data in Figure S13 were filtered to exclude the transcriptionally off state, and are thus not the quantity the current authors are interested in. If they plot the published data for no homie, they will see the clear effect on the average 3D distance, R(s), and a somewhat stronger effect on the contact frequency P(s), which causes significant deviation from the trend-line followed by the homie-containing data.

(2) The agreement between the "best performing" simulations for all models and the Hi-C data is not on par with prior studies using similar approaches, apparently due to some erroneous choices in how the optimization is carried out:

Hi-C-comparison

The 'best fit' simulation Hi-C looks strikingly different from the biological data in all comparisons, with clearly lower agreement than other authors have shown using highly similar methods (e.g., Shi and Thirumalai 2023; Di Pierro et al. 2017; Nuebler et al. 2018; Esposito et al. 2022; Conte et al. 2022), among many others. I believe this results from a few issues with how the current authors select and evaluate the data in their work:

(a) Most works have used Pearson's correlation rather than Spearman's correlation when comparing simulation and Hi-C contact frequencies. Pearson's correlation is more appropriate when we expect the values to be linearly related, which they should be in this case, as they are constructed indeed to be measuring the same thing (contact frequency), just derived from two different methods. Spearman's correlation would have been justifiable for comparing how transcription output correlates with contact frequency. This may fix the bafflingly low correlations reported at lower adhesion values in Figure S2C.

(b) Choice of adhesion strengths - The Hi-C map comparison in Figure 3 strongly suggests that a much more striking visual agreement would have been achieved if much weaker (but still non-zero) homotypic monomer affinity had been selected. In the authors' simulation, the monomer state (A/B identity) strongly dominates polymer position, resulting in the visual appearance of an almost black-and-white checkerboard. The data, meanwhile, look like a weak checkerboard superimposed on the polymer.

(c) A further confounding problem is the aforementioned issue that the Hi-C data don't come from the edited cell lines, and that the interaction of the two Homie sites is vastly stronger than the compartment interactions of this region of the genome.

(3) Some important concepts from the field are ignored:

The crumpled/fractal globule model is widely discussed in the literature (including the work containing the data used in this study) - its exclusion from this analysis thus appears as a substantial gap/oversight:

A natural alternative to the much-discussed Rouse polymer model is the "crumpled polymer" (Grosberg et al. 1988; Grosberg 2016; Halverson et al. 2011; Halverson et al. 2011), also known as the "fractal globule" (Lieberman-Aiden et al. 2009; Mirny 2011; Dekker and Mirny 2016; Boettiger et al. 2016), much discussed for the way it captures the ⅓ scaling of R(s), found for much of the genome (or, equivalently, the -1 exponent of the probability of contact as a function of genome separation, P(s)). Given the 1/3rd scaling in the data, and the fact that the original authors highlighted the crumpled model in addition to the Rouse model, it seems that this comparison would be instructive and the lack of discussion an oversight. Moreover, while prior works (e.g., Buckner, Gregor, 2023) used some traditional simplifying assumptions to estimate the MSD and relaxation time scaling of this model, I believe a more rigorous analysis with explicit simulations (as in Figure 1 for the Rouse model) would be instructive for the crumpled polymer simulations. Note the crumpled globule is not necessarily the same as the globule in the coil-globule transition discussed here - it requires some assumptions about non-entanglement to stay trapped in the meta-stable state which has the 1/3rd R(s) scaling that is indicative of this model, and not the 1/2 exhibited by equilibrium globules (for s<< length of the polymer) and dilute polymers alike.

While the fit in Figure 2 appears to get closer to the 1/3rd exponent (B= 0.32), this appears to be a largely coincidental allusion of agreement - the simulation data in truth shows a systematic deviation, returning to the 1/2 scaling for distances from 500 kb to whole chromosomes. This feature is not very evident as the authors restrict the analysis to only the few points available in the experimental data, though had they tested intervening distances I expect they would show log-log P(s) is nonlinear (non-powerlaw) for distances less than the typical loop length up to a few fold larger than the loop length, and thereafter returns to the scaling provided by the 'base' polymer behavior. This appears to be Rouse-like in these authors' model, with R(s) going like 1/2, even though the data are closer to 1/3rd, as indeed most published simulated P(s) curves based on loop extrusion - e.g., (Fudenberg et al. 2016; Nuebler et al. 2018). In this vein, it would be instructive to the readers if the authors would include additional predictions from the simulation on the plot that lie at genomic separation distances not tested in the data, to better appreciate the predictions.

Minor issues

(1) I think it is too misleading to only describe the experimental data from Brukner as "E-P" interactions from Drosophila. It is important to note somewhere that this is not an endogenous interaction with a functional role in Drosophila - it is a synthetic interaction between enhancers in the vicinity of the eve gene and a synthetic promoter placed at a variable distance away. The uniformity is elegant - (it is the same pair of elements being studied at all distances), but also provides limited scope for generalization as suggested by the current text. Moreover, the enhancers were not directly labeled; rather, the 3D position of nascent RNA transcribed from eve was tracked with an RNA-binding protein and used as a proxy for the 3D position of the enhancers. There is not an individual enhancer at the eve locus that interacts with the transgene, but rather a collection of enhancers is distributed at different positions throughout the entire TAD, which contains eve, and must form separate loops to reach eve. Indeed, it was previously reported that differences in the local position of these enhancers, relative to eve, affect their ability to interact with the distal reporter gene and the endogenous eve gene (Chen 2018). There is also reported competition between these enhancers and the distal gene, which further complicates the analysis (especially since the state of eve and of its enhancers varies among the different cells as a function of stripe position) - see Chen 2018. All of this is ignored in the current work, despite the assertion of the application to understanding E-P interaction. A detailed discussion of these issues is not necessary, but I fear that ignoring them entirely is to invite further confusion and error.

(2) I believe this sentence is overstated, given available data: " TAD borders are characterized by transitions between epigenetic states rather than by preferentially-bound CTCF [4, 23, 24]." Indeed, this claim has been repeatedly made in the literature as cited here. However, other data clearly demonstrate a strong enrichment of CTCF at TAD borders (and at epigenetic borders, which in Drosophila have a high correspondence with TAD borders, as the authors have already appropriately noted). See, for example, Figure 4 of Sexton Cell 2012, and compare to Figure 2 of Dixon 2012. Of minor note, CTCF peaks co-occupied by the Zinc Finger TF CP190 are more likely to be TAD borders than CTCF alone. How big a species-specific difference this is remains unclear, as it appears some mammalian CTCF-marked TAD boundaries may be co-occupied by additional ZNFs. While plenty of Drosophila TAD boundaries indeed lack CTCF, many are marked by CTCF, this is enriched relative to what would be expected by chance (or relative to the alignment of other TFs, like Twist or Eve with TAD boundaries), and it has been shown that CTCF loss is sufficient to remove a subset of these, see for example Figure 5 of (Kaushal et al. 2021) (though it is possible, most will require mutation of the all the border-associated factors that collectively bind many of the borders, dCTCF, CP190, mod(mdg4) and others).

(3) This assertion is overstated given available data: "Although TAD boundaries in Drosophila are often associated with insulator proteins [20], there is no direct evidence that these elements block LEFs in vivo. Therefore, we did not impose boundary constraints in our simulations; LEFs were allowed to move freely unless stalled by collisions with other LEFs, with the possibility of crossover.". Deletion of insulator in Drosophila that lie within a common epigenetic state leads to fusion of TADs (e.g., Mateo et al., 2019 - deletion of the CTCF-marked Fub insulator, in posterior tissues where both flanks of Fub are active; Kaushal, 2021, has examples as well). Loss of CTCF causes a small number of TADs to fuse as measured by Hi-C. This is far from 'direct evidence that insulators block LEFs' - as the authors have already noted, even the idea that cohesin extrudes loops in Drosophila in the first place is indeed

controversial. However, LEF activity and stalling at insulators would provide a very natural explanation of why chromatin in a shared epigenetic state should form distinct TADs, and why these TADs should fuse upon insulator deletion. Justifying the lack of stalling sites based on empirical data is thus not very convincing to this reviewer. I believe it would be more apt to simply describe this as a simplifying assumption, rather than the above phrase, which may be misleading.

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Author response:

We thank the editors and the reviewers for their constructive comments, which have greatly helped us identify key areas to strengthen the manuscript. We acknowledge the validity of the major points raised, and we plan the following revisions:

Criticality

As suggested by Reviewer #1, we will carefully examine whether the dynamics we observe are indeed poised near criticality. We will perform additional analyses to assess how structural and dynamic features change when parameters are tuned away from the coil–globule transition, and we will revise the title and text to ensure that our claims are appropriately moderated.

Role of the *homie* element

We agree with Reviewer #2 that the presence of *homie* elements introduces major modifications to chromosome structure and dynamics. We initially considered that this factor might even explain the paradox described in Gregor's work. In the first phase of our study, we carried out simulations including *homie* elements and found that the potential confounding effects are largely resolved if we restrict the analysis to trajectories prior to encounters between the two *homie* copies. We will include these simulations and expand the discussion accordingly in the revised version.

Comparison to Hi-C data

Both reviewers noted a visual discrepancy between experimental and simulated Hi-C maps. We will address this by testing alternative similarity measures (e.g., Pearson correlation, as suggested) and by exploring parameter ranges that may improve the agreement.
Together, these modifications will strengthen the manuscript, clarify the scope of our conclusions, and directly address the reviewers' central concerns.

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